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Dear Delegates,

Welcome to the International Atomic Energy Agency committee! The IAEA, established in 1957 under the auspices of the United Nations, serves as the world's central intergovernmental forum for scientific and technical cooperation in the peaceful use of nuclear technology. With 178 member states, the Agency promotes the safe, secure, and peaceful application of nuclear technology while working to prevent its misuse for military purposes. This committee will tackle two of the most pressing challenges at the intersection of nuclear security and energy policy. The first topic—addressing undeclared nuclear facilities and clandestine enrichment programs—will ask delegates to confront the hard realities of nuclear nonproliferation, where state deception and the limits of international inspection regimes threaten global security. The second topic—enhancing the safety, regulation, and public acceptance of nuclear power plants—turns to the civilian side of the nuclear equation, where aging infrastructure, inconsistent regulatory standards, and deep public skepticism complicate nuclear energy's potential role in the clean energy transition. Together, these topics represent the dual mandate that has defined the IAEA since its founding: preventing the worst while enabling the best that nuclear technology has to offer. We look forward to a rigorous and rewarding weekend of debate!

Shanna Chung and Justin Kim

Co-Chairs, IAEA, 2025

Topic A: Addressing Undeclared Nuclear Facilities and Clandestine Enrichment Programs

Background

The proliferation of undeclared nuclear facilities and clandestine enrichment programs represents one of the most pressing threats to global security and the international non-proliferation regime. Since the creation of the Nuclear Non-Proliferation Treaty (NPT) in 1968, the international community has worked to prevent the spread of nuclear weapons technology. However, several state and non-state actors have circumvented oversight mechanisms to pursue covert nuclear capabilities, undermining the authority of the International Atomic Energy Agency (IAEA) and the broader framework of international law.

The IAEA serves as the world's primary nuclear watchdog, mandated to verify that member states are using nuclear materials and technology solely for peaceful purposes. Through its safeguards system — a set of technical measures including inspections, environmental sampling, and remote monitoring — the Agency seeks to provide credible assurances that nuclear material is not being diverted to weapons programs. Despite these mechanisms, historical cases such as Iraq's pre-1991 weapons program, North Korea's repeated violations of safeguards agreements, Iran's concealed enrichment facilities, and Syria's suspected undeclared reactor at Al-Kibar have exposed significant gaps in the international community's ability to detect clandestine activities in a timely manner.

Clandestine enrichment programs pose a unique challenge because uranium enrichment technology is inherently dual-use: the same centrifuge cascades that produce low-enriched uranium for civilian reactor fuel can, with further operation, produce highly enriched uranium suitable for weapons. States have exploited this ambiguity to develop latent nuclear capabilities under the guise of civilian energy programs, making it extraordinarily difficult to distinguish

peaceful intent from weapons development before a critical threshold is crossed. The international community must therefore grapple not only with detecting undeclared facilities, but with developing the legal and diplomatic tools necessary to respond effectively once violations are discovered.

Relevant Issues

A central challenge in addressing undeclared nuclear programs is the limitation of the IAEA's legal authority to conduct intrusive inspections. Under the standard Comprehensive Safeguards Agreement (CSA), the IAEA can only inspect declared facilities and nuclear materials. The Additional Protocol (AP), which grants inspectors broader access rights including the ability to visit undeclared sites, is voluntary and has not been adopted by all NPT member states. This gap allows states to conceal nuclear activities in facilities that have never been reported to the IAEA, rendering the standard safeguards system effectively blind to clandestine programs that operate entirely outside declared channels.

Intelligence sharing between member states and the IAEA presents another systemic vulnerability. The Agency itself has limited independent intelligence-gathering capabilities and relies heavily on information provided by member states — many of which are reluctant to share sensitive intelligence for political or strategic reasons. This creates dangerous asymmetries of information that can delay the detection of violations by years. The cases of North Korea and Iran demonstrate how significant technical progress can be achieved before international pressure can be applied, particularly when a state strategically manages the pace and scope of its disclosures to the Agency.

The geopolitical dimension of non-proliferation further complicates enforcement. The UN Security Council, which holds ultimate authority to impose binding measures on states that violate their safeguards obligations, is frequently paralyzed by the vetoes of permanent members with divergent strategic interests. States suspected of clandestine nuclear activities can exploit these divisions to escape meaningful consequences, weakening the deterrent effect of international norms. Furthermore, states possessing nuclear weapons outside the NPT framework — including India, Pakistan, and Israel — create a credibility gap in the non-proliferation regime that other states have historically cited to justify their own pursuit of nuclear capabilities.

Possible Solutions

Strengthening the universality of the Additional Protocol is a critical first step toward closing existing verification gaps. Delegates should consider mechanisms for incentivizing or requiring all NPT member states to conclude and bring into force an Additional Protocol, potentially linking AP membership to access to civilian nuclear cooperation agreements, fuel supply guarantees, or other tangible benefits. Mandatory adoption could also be pursued through a new UN Security Council resolution modeled after Resolution 1540, which required all states to adopt domestic measures against the proliferation of weapons of mass destruction by non-state actors.

Technological innovation offers promising avenues for improving detection capabilities. Advances in satellite imagery analysis, artificial intelligence, and open-source intelligence (OSINT) now allow independent analysts and governments to identify signatures of undeclared nuclear activities — such as unusual construction patterns, water intake systems consistent with reactor cooling, or specialized procurement patterns — with growing precision. The IAEA

should be empowered and adequately resourced to leverage these technologies systematically, including through enhanced partnerships with academic institutions and commercial satellite providers. Expanded environmental sampling programs can furthermore detect trace isotopes indicative of enrichment activities across wide geographic areas without requiring direct physical access.

Diplomatically, the international community must develop more graduated and credible response frameworks for states found to be in non-compliance. A tiered response system, agreed in advance by the international community, that automatically triggers escalating consequences proportionate to the severity and duration of a violation could significantly improve deterrence. Additionally, regional denuclearization agreements and the expansion of nuclear-weapon-free zones represent important confidence-building measures that can reduce the underlying incentives for states to pursue clandestine capabilities, particularly in regions where security competition and threat perception are primary drivers of proliferation behavior.

Questions to Consider:

1. How can the IAEA's legal authority and financial resources be strengthened to enable more proactive and intrusive inspections of suspected undeclared nuclear facilities?
2. What mechanisms can incentivize universal adoption of the Additional Protocol, and how should the international community respond when states refuse to grant inspectors access to suspected sites?
3. How should the international community balance the sovereign right of states to develop peaceful nuclear energy with the necessity of intrusive verification measures to prevent weapons proliferation?

4. What role can emerging technologies such as artificial intelligence, remote sensing, and environmental sampling play in detecting clandestine enrichment activities, and how should their use be governed internationally?
5. How can the Security Council reform its decision-making processes to respond more swiftly and effectively to cases of nuclear non-compliance without being paralyzed by geopolitical divisions among permanent members?

Helpful Links:

- 1) <https://www.iaea.org/topics/safeguards-and-verification>
- 2) <https://disarmament.unoda.org/en/our-work/disarmament-bodies/disarmament-fellowship>
- 3) <https://www.armscontrol.org/>



Topic B: Enhancing the Safety, Regulation, and Public Acceptance of Nuclear Power Plants

Background

Nuclear power currently supplies approximately 10% of the world's electricity, generated by around 440 operational reactors across 32 countries. As the global community grapples with the urgent need to decarbonize energy systems and reduce greenhouse gas emissions, nuclear power has re-emerged as a significant point of debate in climate and energy policy discussions. Unlike fossil fuels, nuclear energy produces virtually no direct carbon emissions during operation, making it an attractive option for countries seeking to meet both growing energy demands and international climate commitments under agreements like the Paris Accord.

The history of nuclear power is, however, deeply intertwined with some of the most consequential industrial accidents of the modern era. The 1979 partial meltdown at Three Mile Island in the United States, the 1986 catastrophic explosion at Chernobyl in the Soviet Union, and the 2011 Fukushima Daiichi disaster in Japan—triggered by a tsunami—each fundamentally reshaped public attitudes toward nuclear energy and prompted sweeping revisions to safety standards and regulatory frameworks worldwide. These events demonstrated that even well-resourced nations with established nuclear programs were vulnerable to failures arising from a combination of technical malfunction, human error, and inadequate regulatory oversight.

The IAEA, founded in 1957 under the auspices of the United Nations, serves as the world's central intergovernmental forum for scientific and technical cooperation in the peaceful use of nuclear technology. Its mandate includes establishing safety standards, conducting peer reviews of member states' nuclear facilities, and providing technical assistance to countries developing or expanding nuclear infrastructure. However, IAEA safety standards are not legally

binding—their implementation depends entirely on the domestic legislation of each member state—creating significant disparities in how nuclear safety is enforced globally.

As of 2025, a new generation of reactor designs, including small modular reactors (SMRs) and advanced Generation IV reactors, is entering the policy conversation. These technologies promise improved passive safety features and reduced waste output, but they also raise novel regulatory questions that existing frameworks were not designed to address. At the same time, a growing number of developing nations are pursuing nuclear energy for the first time, introducing new variables in terms of regulatory capacity, workforce training, and emergency preparedness. The international community thus faces a dual challenge: maintaining rigorous oversight of existing plants while creating adaptive frameworks capable of governing the next era of emerging nuclear technology.

Relevant Issues

One of the most fundamental concerns is the inconsistency of regulatory standards across IAEA member states. While countries like France, the United States, and South Korea have well-established independent regulatory bodies and decades of operational experience, many nations pursuing new nuclear programs lack the institutional infrastructure to enforce safety standards effectively. Regulatory capture—the phenomenon by which oversight bodies come to prioritize the interests of the industry they regulate over public safety—remains a structural vulnerability in jurisdictions where nuclear operators and regulators are not sufficiently independent from one another.

Waste management represents another critical and politically contentious dimension of the nuclear safety debate. Despite decades of research, no country has yet successfully opened a permanent deep geological repository for high-level nuclear waste, which remains radioactive for

thousands of years. In the interim, spent fuel is typically stored in cooling pools or dry casks at reactor sites—a solution that was intended to be temporary but has become de facto permanent in many countries. The accumulation of waste at plant sites presents both a safety risk and a major obstacle to public acceptance.

The human dimension of nuclear safety is equally significant. Aging reactor fleets in Europe and North America are being operated beyond their original design lifespans, raising questions about the integrity of aging infrastructure and the adequacy of maintenance protocols. The increasing complexity of cybersecurity threats adds a further layer of vulnerability, as modern nuclear control systems have become more digitized and therefore potentially susceptible to interference. Public perception compounds all of these challenges: studies consistently show that public trust in nuclear power is low in many countries, shaped by memories of major accidents, concerns about waste, and distrust of regulatory institutions. This skepticism has had direct policy consequences, as seen when Germany accelerated its nuclear phase-out following Fukushima. The gap between expert risk assessment and public risk perception represents a true governance challenge.

Possible Solutions

Given the breadth and complexity of nuclear safety challenges, effective solutions must address technical, institutional, and social dimensions simultaneously. Strengthening the IAEA's authority is a logical starting point. While binding international safety conventions like the Convention on Nuclear Safety already exist, their review mechanisms lack enforcement power. Delegates might consider how peer review processes such as the IAEA's Operational Safety Review Teams (OSART) could be made more systematic and their findings more consequential, without infringing on national sovereignty over energy policy.

Harmonizing regulatory standards across member states—particularly for emerging nuclear nations—represents another significant opportunity. On waste, sharing best practices on geological disposal and potentially developing shared regional repository solutions could break the current policy deadlock. Transparent public engagement processes, in which affected communities are given meaningful input into siting decisions, have shown promise in Finland and Sweden, where site selection has proceeded with relatively broad public support.

Finally, rebuilding public trust demands sustained, honest, and two-way communication. Transparency about past accidents and regulatory failures—rather than defensive minimization—has consistently been shown to build more durable public confidence than reassurance alone, and should be treated as a core component of any credible nuclear governance strategy.

Questions to Consider:

- 1) How can the IAEA strengthen its role in enforcing international nuclear safety standards while respecting member states' sovereignty over their domestic energy policy?
- 2) What mechanisms can be established to ensure that countries newly pursuing nuclear energy develop sufficient regulatory capacity before their first plants become operational?
- 3) How should the international community address the long-term storage and disposal of nuclear waste, particularly in countries that lack the resources or geography for a domestic repository?
- 4) What role should public engagement and participatory governance play in nuclear licensing and oversight processes, and how can trust between nuclear institutions and host communities be rebuilt?

- 5) How can regulatory frameworks be updated to safely accommodate new reactor technologies like small modular reactors, without creating duplicative national review processes that slow deployment?

Helpful Links:

- 1) <https://www.iaea.org/topics/energy>
- 2) <https://www.iaea.org/newscenter/news/what-is-nuclear-energy-the-science-of-nuclear-power>
- 3) <https://energy.mit.edu/research/future-nuclear-power/>



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